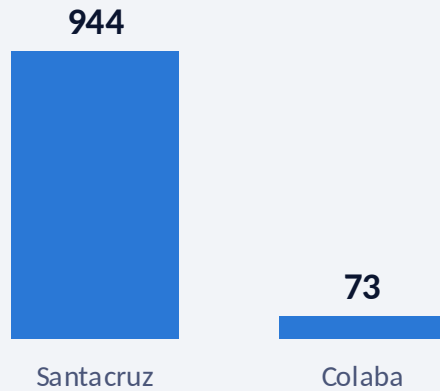


A weather-decision problem, and a signal Jio may already carry

SAME CITY, DIFFERENT WEATHER

Rainfall (mm) • 26 July 2005 • IMD



WHAT A CITY-LEVEL ALERT SAYS

“Heavy rain in Mumbai.”



WHAT IT NEEDS TO DECIDE

“Which zones do I act on, right now?”

Stations ~20 km apart. In 2020–24, on 5 of 15 heavy-rain days the other station got less than half.



Businesses decide blind
dispatch, incentives, delivery promises



Workers carry the loss
income and safety, street by street

India doesn’t have a weather-data problem.
It has a weather-decision problem.

THE INSIGHT: JIO’S MICROWAVE LINKS MAY CARRY A WEATHER SIGNAL



Rain weakens the signal. Measure the excess loss and estimate path-averaged rainfall along each link, in near-real-time.



Microwave links: usable



Fibre: no rain signal

JIOMASAM IS

decision infrastructure, not a weather map

Link signals + IMD radar
+ rain gauges



Spatial weather grid



Decision-ready alerts
and triggers

IT IS

- Weather intelligence from existing infrastructure
- Alerts and data for businesses
- A trigger for future worker protection

IT ISN'T

- A new sensor network
- A consumer weather app
- An insurer, or heat sensing from towers

Links alone fall short of our bar, so accuracy decides the mix

✓ **PROVEN • PUBLISHED** Physics: rain weakens microwave signals. Method: Science 2006, PNAS 2013 (country-wide maps).

🔪 **OUR CHECK • PUBLIC DATA** We ran the open-source method on 500 real links (Germany), radar as reference. Not Jio data.

⌚ **NOT YET PROVEN • THE PILOT** Jio's coverage, Indian accuracy, business value, who pays, insurance-grade triggers.

68%

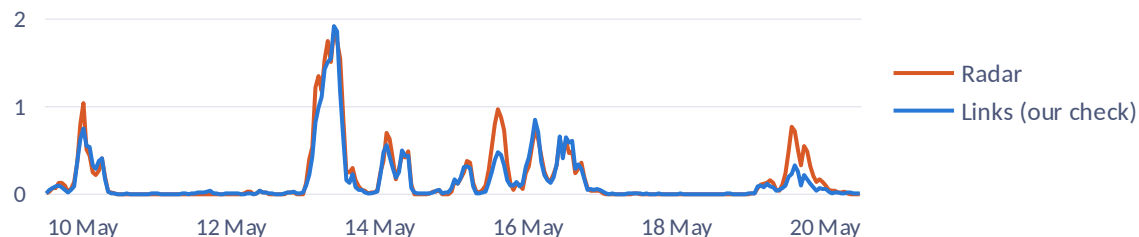
of rain events caught by links alone
1,120 of 1,639 radar events • 8 unseen test days

links alone: 68%

our bar: 80%

Event = 10 mm+ of rain in 3 hours in a ~4 km zone. **Below our bar, so we never use links alone.**

Network average, hourly (mm/h): links tracked radar closely



CANDIDATE SOURCES AND WHO PROVIDES THEM

“” Jio microwave links	Jio-owned
✂ Network health: faults, outages, maintenance	Jio-owned
📡 Doppler radar: spatial check	IMD (govt)
🕒 Rain gauges: ground truth	IMD • BMC
✂ Satellite: coverage fallback	External
📍 New gauges on Jio sites, only where links are thin	New install
⚖ Commercial data, only if it lifts accuracy	Commercial

THE PILOT SCORES EVERY MIX AGAINST HELD-OUT RAIN GAUGES

Links only

our check: 68%

Links + radar + gauges

measured in the pilot

Radar + gauges, no links

shows what Jio adds

Rule: the lowest-cost mix that clears every gate: ≥80% of trigger events • totals within ±10% • under 15% unconfirmed

Prove it inside Reliance, then sell the same intelligence

1 • FIRST PROVING GROUND: RELIANCE



THE HERO TEST
Jio network operations

- Can we tell rain-related signal loss from equipment faults before sending a crew?
- Potentially fewer wasted visits and earlier flood alerts
- Jio owns the network data and controls the decision

Tests accuracy and savings



JioMart and Reliance Retail
Rain-aware dispatch, delivery promises, zone pauses
Tests commercial value



MyJio
Street-level rain alerts for users
Tests reach at scale

2 • THEN SELL THE SAME INTELLIGENCE



Rain hits Zone A



JioMausam sends a zone alert + rain estimate



The customer decides: reroute, update ETAs, adjust incentives



Lower disruption

WHO BUYS, AND WHAT

1	Delivery and quick commerce	Alerts
2	Logistics, cities, insurers	Alerts + data
3	Agriculture, power, retail	Data + alerts

3 • WHY JIO: INTEGRATION, NOT THE SENSOR



SENSE Jio network



UNDERSTAND JioMausam



REACH Reliance ecosystem



PROTECT JFS + licensed insurers (optional)

HOW JIO CREATES VALUE

01 Intelligence fees External: API, alerts and data

02 Savings inside Reliance Internal: network ops, delivery, retail

03 Distribution fees Optional: only if Kavach launches

The business stands even if insurance never launches.

Every internal use is a hypothesis the pilot tests.

We sell: zone alerts • a live rain feed (API) • auditable rain records • trigger data for insurers. Not a weather map.

Insurance comes last. Test the rest in 90 days

WHERE WE STARTED: MAUSAM KAVACH, ONE APPLICATION, NOT THE ANCHOR



Why optional: only after the accuracy and actuarial gates (shadow payouts until then) • who pays is still unproven • the insurer carries the risk, never Jio • if it fails, nothing breaks. Eligible = already active in the zone before the event began.

IF WE OFFER IT, WEATHER RISK SETS THE STARTING PRICE

City • product	Trigger	Days a year	Paid days	₹ a month
Mumbai • Basic	≥204.5 mm rain a day	1.1	1.1	₹45
Nagpur • Basic	≥47°C	2.2	~1.2	₹52
Ahmedabad • Basic	≥47°C	0.7	~0.6	₹29 (min)
Mumbai • Plus	≥64.5 mm rain a day	11.9	5 (cap)	₹208

Price = expected paid days × ₹300 ÷ 60% ÷ 12
25 years of IMD records (2000–24). ₹300 and 60% are prototype assumptions; final pricing after actuarial validation. One flat price can’t fit India’s risks.

THE ASK: THREE GATES IN 90 DAYS

DAYS 1–30
Coverage

Audit Jio’s microwave links and data readiness; the audit picks the live city
Gate 1: enough usable coverage?

DAYS 15–60
Accuracy

Back-test archived Mumbai monsoon data against held-out IMD and BMC gauges
Gate 2: does the fused system clear every gate?

DAYS 61–90
Value

Live shadow test in one Reliance operation, matched zones with vs without
Gate 3: does it improve a real operation?

Only after all three gates pass do we scale externally. If not, we stop.

	0–3 months	4–12 months	Year 2	Year 3
Intelligence	Three gates	Reliance live + first clients	Cities + enterprises	National weather API
Kavach (optional)	Shadow payouts	Sandbox + heat	Basic at recharge	National scale

We aren’t asking Jio to build another weather network, just to discover what its existing one can tell us.
First prove the signal. Then sell the intelligence. Protection comes last.